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# KINAITICS: Enhancing Cybersecurity Education Using AI-Based Tools and Gamification

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## Abstract

In the vanguard of the digital age, cybersecurity education emerges as a pivotal force in the ever-evolving threat landscape that challenges individuals, organizations, and nations alike. Cybersecurity education serves as the foundational layer that empowers stakeholders to understand, anticipate, and mitigate potential cyber threats. Furthermore, in an era where artificial intelligence systems are ubiquitous, they can be both the target and the tools of sophisticated cyberattacks. This paper presents a comprehensive narrative that intertwines the foundational importance of cybersecurity education with the application of gamification strategies and the creation of a cyber range for hosting real-life use cases and AI-based systems related to several domains aligned with a European project. This work meticulously documents the execution and outcomes of two hackathon Capture-The-Flag events, offering a comprehensive analysis of participant engagement, performance metrics, and the invaluable lessons distilled from these immersive experiences. The synthesis of theoretical knowledge with real-world application encapsulated in these events epitomizes the essence of effective cybersecurity training, charting a course for future endeavours in the domain.

## CCS Concepts

• **Social and professional topics** → **Adult education; Software engineering education;** • **Security and privacy** → **Software security engineering;**

## Keywords

Education, Training, Cybersecurity, Hacking

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## 1 Introduction

Artificial Intelligence (AI)-enabled systems are increasingly adopted in several domains. Although AI systems offer several advantages, such as task automation and information extraction in an era where IoT systems produce vast amounts of data, they are also new targets of adversarial attacks, which could trick these systems and make them more harmful than beneficial. According to a recent study from HiddenLayer[16], AI security breaches are becoming a serious problem in the industry. The survey showed that 77% of businesses reported a breach in their AI systems in the last year. Similarly, the Zscaler 2024 security report[28] indicates that 18.5% of all AI enterprise transactions are blocked due to security concerns. Additionally, AI is being weaponized for malicious purposes by adversaries, enhancing and fine-tuning their attacks[18]. Research findings show that existing cyber defence infrastructures will become inadequate to address the increasing speed and complex decision logic of AI-driven attacks[3]. Thus, organisations that adopt such systems need to be informed about the dangers they introduce to reap their benefits and avoid any pitfalls with severe consequences. Furthermore, according to Verizon's 2024 Report[25], 68% of breaches involve a non-malicious human element, such as a person falling victim to a social engineering attack or making an error.

For this reason, several publications highlight that one of the key factors for preventing cyberattacks is user awareness and training [6] [5]. Cybersecurity training is crucial for educating people about the various attack vectors that can threaten AI-based systems and raise awareness. It's essential for companies to allocate resources to cybersecurity education and training initiatives. These programs are crucial in informing staff about the potential dangers associated with AI-driven systems and fostering an environment where security is a shared responsibility. Educating the workforce

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on prevalent methods of attack, the nuances of social engineering, and the fundamentals of maintaining digital cleanliness can equip them with the knowledge to identify and alert authorities about any unusual activities. This proactive approach significantly diminishes the chances of cyberattacks being effective [12].

One of the most widely used techniques for educating people about cybersecurity threats is the gamification approach. Research has shown that this method, used as a supplementary tool in cybersecurity training, leads to learning gains, fosters teamwork, keeps students interested in the course material, and encourages participation[1]. In line with this approach, this paper presents our training experiences and findings within the context of the KINAITICS H2020 project <sup>1</sup>. KINAITICS aims to explore new attack opportunities offered by the introduction of AI-based systems, as well as offer an innovative spectrum of tools and methodologies to protect against these threats.

More specifically, the tools developed in the KINAITICS project, along with its Use Cases (UCs), are virtualised within a cyber range. This cyber range is then utilised to train individuals and organisations in cybersecurity skills, assess security tools, and develop effective strategies for defending against cyber threats. The training is conducted through dedicated events based on the Capture-The-Flag approach, known as hackathons. This paper provides a detailed description of the infrastructure used in two hackathon events, discusses decisions made to enhance engagement levels, evaluates their impact on both events, and compares outcomes in terms of participants' expertise, progress, and feedback.

These outcomes serve as validation for the functionality of the attack and defence tools developed as part of KINAITICS. Furthermore, post-project, the cyber range that supported the hackathon challenges will continue to serve as educational material, enhancing the impact of the project results and raising awareness about new cybersecurity threats.

The rest paper is organised as follows: Section 2 presents the learning approaches and environments traditionally adopted for cybersecurity education, as well as current trends in the domain, and subsequently briefly describes the use cases (UCs) of KINAITICS project. Then, Section 3 refers to the infrastructure that supported the KINAITICS hackathons, including the Capture-The-Flag platform, cyber range architecture, challenge configuration, and methodology used for evaluating the events. Section 4 reports the hackathon agendas, and the gathered results in terms of participants' expertise, their progress level and their feedback, as well as our lessons learned. Finally, conclusions and future work are described in Section 5.

## 2 Background

### 2.1 Learning Approaches and Environment

In the evolving landscape of cybersecurity education, there are many different training approaches and techniques, others are more conventional based on pre-built systems and fixed curricula and others utilise more modern approaches such as context-aware training systems and virtual and augmented reality (VR/AR) assisted training. Conventional approaches include the usage of text material, videos, online training, cyber ranges and virtual environments.

<sup>1</sup><https://kinaitics.eu/>

On the other hand, modern approaches may rely on user-specific characteristics, where the training material is tailored to the needs and skills of certain users or use technologies such as VR/AR to provide an immersive and realistic training experience [2]. The incorporation of the AI and Extended Reality technologies can bolster both the effectiveness and efficiency of education. These emerging trends, by providing personalised, engaging, and real-time training experiences, have the potential to be the foundation of modern cybersecurity education strategies [23]. However, conventional methods are still predominantly used, as they offer a solid foundation for cybersecurity training. Among these methods, the gamification approach is widely applied for integrating game design elements into non-game contexts and to engage and motivate people to achieve their goals. The gamification is usually supported by deploying a battlefield in a cyber range, which is a virtual environment designed for cybersecurity training, testing, and research.

Once the gamification approach is chosen and the cyber range is deployed, dedicated event called hackathons are organised. These intensive events harness collaborative problem-solving to tackle real-world cybersecurity challenges. Such live activities improve applied learning performance compared to conventional classroom/lab study combined with simulations [26, 27]. Current trends in hackathon design integrate traditional, technology-based, and innovative training methods to cultivate a comprehensive skill set among participants. Moreover, hackathons often incorporate Capture-the-Flag (CTF) competitions, which simulate cybersecurity scenarios that require participants to employ offensive and defensive strategies in real-time[24]. These practices not only enhance technical proficiency but also foster soft skills such as teamwork, communication, and critical thinking [14]. CTF competitions follow mainly two formats, Jeopardy-style and Attack/Defence style [11]. In Jeopardy-style CTFs participants can choose between a set of challenges each of which is assigned a value representing the points that will be gained if the challenge is successfully solved. In the Attack/Defence style, each team of participants controls an identical instance of a computer network, which includes systems with vulnerabilities. The teams try to protect their network assets and at the same time attack the networks of other teams. There are also some other not-so-common formats found in the literature, such as King of the Hill [4], where the main goal is capturing a target system and keeping it under control for as long as possible and mixed format which is a combination of Jeopardy and Attack/Defence formats.

The lack of Attack/Defence team specialisation, the presence of multiple users, the high number of tools and their different domains of application, as well as the various project use cases (see Section 2.2), led us to choose the Jeopardy-style format to create a competitive environment and engage multiple users simultaneously.

### 2.2 Project Scenarios

KINAITICS attack and defence technologies will be aligned and verified on 6 use cases (UCs) defined within the project. However, currently, tools have been used and tested only in 5 UCs, which are the following:

**2.2.1 UC1: Attacks on simulation code to design critical facilities.** This use case addresses the effects of attacks against an uncertainty quantification calculation platform based on inputs and outputs of a simulation code for nuclear facilities. The uncertainty quantification platform assess how system inputs can influence the outputs and the behaviour of an energy system. The main attacks under investigation are data tampering attacks, either with the inputs or outputs of the simulation code which could lead to false conclusions of the uncertainty quantification platform and therefore erroneous design of critical systems. Another potential attack with less importance is the parameter inference attack, where the attacker, having access to large amounts of input and output data, can conclude the simulation code.

**2.2.2 UC2: Phishing email to steal EHR data.** Phishing e-mails are a very common attack vector weaponized by attackers to gain a foothold in a private network. In this particular use case, a Hospital Information System (HIS) is the final attack target and more specifically the Electronic Healthcare Record (EHR) system that manages patient healthcare data. The attacker launches a phishing e-mail campaign targeting users of the EHR system such as doctors, nurses or technicians, aiming to lure them to expose their credentials. In case of successful credential harvesting, the attacker is able to change the credentials, lock the user out of the system and access healthcare data.

**2.2.3 UC3: Image Tampering in Healthcare.** Another potential attack against the HIS under investigation is healthcare image tampering and adversarial attacks on the classification model. This scenario starts once the training process is completed, and the deep learning algorithm can be put into inference mode for new patients. On the one hand (tampering image), the main objective of the adversary is to gain access to the network between the Picture Archiving and Communication System (PACS) server and the client with the intention of tampering with the data. His/her objective is twofold: first, to alter the AI model's learning process by poisoning the data (introducing false information) and adding triggers for backdoor attacks and second, to mislead the model during the inference phase by exploiting the backdoors. On the other hand, the adversary can directly attack the model by slightly modifying the image and so mislead the classification [9].

**2.2.4 UC4: Bot Detection.** Nowadays malicious bots represent an increasing threat to publicly reachable web applications and services. Imperva Bad Bot Report 2024 [15] states that nearly half of all the internet traffic for year 2023 originated from bots. In addition, bad bots accounted for 32% of all internet traffic – a 1.8% increase from 2022. This scenario aims to leverage AI consistently to enhance the protection of Web Applications and Services, by strengthening the bad bot detection performance, while also ensuring that the AI layer empowering the detection of the attacks is resilient against tailored adversarial attacks and generally trustworthy.

**2.2.5 UC5: Structural Health Monitoring.** Structural Health Monitoring (SHM) refers to a Cyber Physical System (CPS) responsible for monitoring or tracking the behaviour of a structure, as well as its overall evolution over time by measuring the response signal to an excitation periodically propagated within it. The aim is to identify defects in the structure by detecting alterations in the signal. This

use case studies the application of SHM to a railway network. The SHM is composed of 3 layers, the physical layer responsible for the emission and reception of signals based on a network of sensors, the communication layer composed of trackside cabinets that gather the information sent by the sensors corresponding to the interface between the sensors and the remote server and the decision layer with the remote server that hosts the software enabling fracture detection based on the information sent by the trackside cabinets. An attacker can either tamper with the sensors' measurements at the physical layer, attack the communication layer to alter the transmitted information from the trackside cabinets to the server or attack the fracture detection AI model to cause false positive or false negative decisions. False positive decisions may have a financial impact, while false negative decisions may lead to catastrophic railway accidents.

### 3 Infrastructure and Hackathon Design

In this section, information about the tools, the design and the structure of the hackathon events are presented.

#### 3.1 Developed Tools

Tools have been developed to cover the issues detected in the project UCs and introduced in Section 2.2. In this sense, it should be noted that some tools are designed to be used across multiple UCs with specified parameterisation and adjustments, while others may only cover specific aspects of a UC due to their specialised focus. Furthermore, tools have different maturity levels, some of them are already ready to be commercialised, while others are in the research stage. At this stage, 11 tools have been developed:

**1. URAttack:** is a multi-platform attack tool that exploits simulation vulnerabilities in a nuclear context using simulation platforms such as Uranie [13]. Specifically, it can exploit two attack vectors: injecting outlier values during data transmission to an external simulation code, and model stealing through surveillance and malicious extraction of the behaviour of a black-box code simulation. This tool is related to UC1.

**2. Social Engineering Detection (SED):** is a tool to assess emails for identifying phishing attacks. In particular, it analyses the header, body, and attachments to detect and explain why the email under assessment is (or could be) malicious. This tool is related to UC2.

**3. PhiShield:** offers a complementary approach to SED in detecting phishing attacks. It employs a hybrid methodology utilising machine learning models [21] to classify HTML and URL information as either phishing or legitimate. One of the main advantages of PhiShield is its use of automatically extracted features directly within the machine-learning model. Additionally, due to its relationship with the SED tool, future steps involve integrating these modules to enhance phishing threat comprehension. This module is tested in UC2.

**4. Multidimensional Data Processing and Intelligent framework (MDPI):** aims to alert and help users recognise cybersecurity threats, make decisions, and take actions rapidly to prevent incidents. The strength of this tool lies in its combination of attributes from the NIST framework [20] and the Joint Research

Center (JRC) taxonomy [19]. This module is tested for analysing and relating events and information related to UC2.

**5. NeuralSentinel:** assesses the reliability of Artificial Neural Network (ANN) models by implementing advanced deep learning attack and defence strategies [10]. Furthermore, the tool uses explainability strategies [8] to increase the trustworthiness of the results. This tool is mainly used in UC3, for assessing the reliability of a system able to classify medical images.

**6. Hybrid Defence System (HDS):** is a tool able to detect and explain threats found in medical training datasets and propose actionable countermeasures. Unlike NeuralSentinel which operates on the model, HDS operates in the dataset trying to detect poisoned or tampered samples. Thus, it applies explainable AI (XAI) to mark interested areas of the images in a human-understandable manner. Finally, human feedback is included in the tool to trigger a Machine Unlearning algorithm [7] and so reduce the impact of poisonous samples from the AI models. This tool is mainly used in UC3.

**7. SeerBox:** is a web application security manager that can be used for monitoring and protecting web applications and services<sup>2</sup>. Leveraging the power of machine learning, its proprietary detection engine is resilient against adversarial attacks targeting application layer vulnerabilities. Seerbox can monitor multiple traffic sources simultaneously and thus enable effective protection even in environments distributed between private data centres and public clouds. This tool is mainly used for addressing issues related to UC4.

**8. Bot Outlier Tracker (BOT):** is a module prototype to be integrated into the Seerbox product to enhance bot detection. It automatically learns the maximum number of legitimate requests per source IP address within a specific time window and identifies outlier source IP addresses (potential bots). The module also allows for human operator feedback to update learned thresholds. This tool is related to UC4

**9. CyberShield:** suggests the best countermeasures for mitigating the effects of detected cybersecurity incidents. Specifically, it tests users' abilities against an attack system trained using a Reinforcement Learning strategy. This tool is tested in UC4 to support users in taking actions and countermeasures to prevent the spread of bot infections in a network.

**10. PenTestFox:** is an offensive security tool designed for reconnaissance, vulnerability discovery, and automatic cyber-attack execution. It operates as a sophisticated cyber agent, navigating the complex digital terrain of a target network. Its primary mission is to gather intelligence, identify weaknesses, and execute calculated attacks that exploit those vulnerabilities. This tool is mainly used in attacking the communication layer of the UC5.

**11. SHM attack tool:** aims to attack AI models designed for identifying defects on rails, a key application within the SHM context. In fact, in SHM, nodes equipped with actuators and sensors are located along the rails, emitting and receiving ultrasound signals to and from nearby nodes. These AI models are trained using ultrasound wave signals to identify patterns indicative of structural defects. This tool is mainly used in attacking the decision layer of the UC5.

These 11 tools are currently being developed under the project, and it was possible to test all of them during the second hackathon. However, in the first event, only four tools, i.e., SeerBox, NeuralSentinel, SED and URAttack, were available.

### 3.2 Cyber Range Infrastructure

Although the KINAITICS project aims to develop and test tools for attacking and defending cyber-physical systems, at this stage, all the tools are designed to operate solely within the cyber environment. For this reason, the cyber range only includes virtualised machines and devices. More specifically, as a cyber range, we utilised a dedicated server equipped with 48 Virtual CPUs, 500GB of RAM, and 18TB of disk storage. On this robust hardware, we have deployed the OpenStack platform<sup>3</sup>, an open-source cloud computing solution renowned for creating and managing extensive networks of virtual private servers. OpenStack's snapshot and deployment processes are exceptionally fast and crucial for the seamless operation of the hackathon. This rapid deployment capability ensures prompt resolution of any disruptions, maintaining the event's flow and minimizing downtime for participants.

Although the number, resources, and configurations of the Virtual Machines (VMs) deployed in the cyber range varied in each event, all of them were divided into three zones: Access, Tools, and Use Cases, as shown in Figure 1. This zone differentiation allows us to regulate and monitor access to the cyber range resources. Specifically, the machines in the Access zone, also referred to as landing hosts, were Ubuntu Desktop 22.04 LTS machines running a Remote Desktop Protocol (RDP) server, enabling participants to access them using an RDP client and utilize the graphical interface of the VM. Simultaneously, firewall ports were opened for each RDP server on the landing host, allowing users to connect from their homes or offices using a combination of public IP addresses and ports, without the need for a VPN connection. Furthermore, to ensure a good experience for the hackathon participants and to optimise the cyber range resources, each landing host was accessed by multiple participants at the same time. However, each one with a different user, allowing the separation of the sessions.

In contrast, the VMs in the other two zones (Tools and Use Cases networks) were accessible only from the landing hosts. More specifically, the Tools zone hosts VMs in which the tools that were used during the hackathon were deployed, while in the Use Cases zone, services, databases, and other utilities needed for recreating the UCs defined in the KINAITICS project (Section 2.2) were deployed. In this case, the Operating System was strictly related to the needs of the tool owners and the UCs' requirements.

**Configurations.** In the first hackathon, 4 landing hosts and 3 VMs for hosting the tools available at the event were deployed, along with 3 additional VMs for hosting UC resources (e-commerce, DICOM database, and mail server). In the second event, the number of landing hosts was increased to 8 in anticipation of more participants and to ensure spare machines were available if needed. Furthermore, to reduce the overloading problems experienced during the first event, each landing host was equipped with a Prometheus

<sup>2</sup><https://seerbox.it/en/>

<sup>3</sup><https://www.openstack.org/>

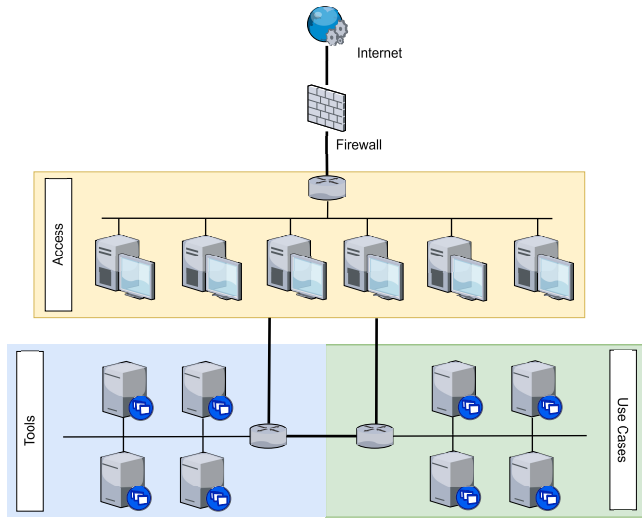


Figure 1: Hackathon network diagram

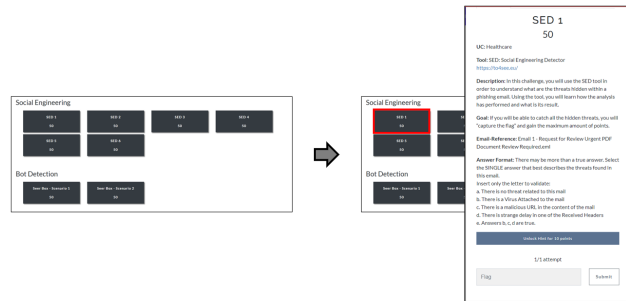


Figure 2: Challenges example

agent<sup>4</sup> for real-time resource monitoring. Additionally, in the second event, 9 VMs were deployed for hosting the available tools and 3 VMs for the UC resources.

### 3.3 Hackathon challenges

The main idea of the hackathon events was to share with the participants and the cybersecurity community, new tools and technology developed in a dedicated KINAITICS project aimed at attacking and defending cyber-physical systems using AI systems. At the same time, these events allow tool owners to gather feedback about the usability, strengths, and weaknesses of their tools, thereby accelerating their development.

In particular, the hackathon events were organised as a Capture-The-Flag (CTF) game following a Jeopardy-style strategy. Figure ?? shows an example of challenge included in the CTF platform during the hackathon events. Moreover, Figure 3 reports the infrastructure and the operations users need to follow when this strategy is implemented. Firstly, participants select a challenge from the board (point 1) and then access the cyber range (point 2), where the tools and UC services are deployed (as presented in Section 3.2).

<sup>4</sup><https://prometheus.io/>

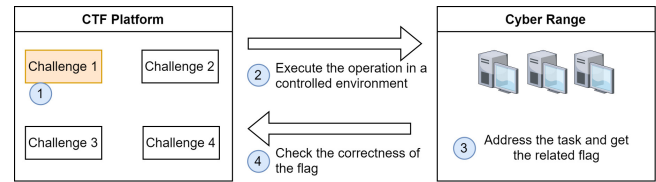


Figure 3: Jeopardy-style actions flow

Here, they execute various actions, operations, or tests to solve the challenge and find the required flag (point 3). Finally, they return to the CTF controller platform to submit their answer (point 4). If the answer is correct, they earn the corresponding points. The objective is to earn as many points as possible before time runs out.

There are several CTF platforms for hosting hackathons in Jeopardy-style. Some of them are open-source, such as CTFd, FacebookCTF, HackTheArch, Mellivora, Pedagogic-CTF, PicoCTF, RootTheBox and WrathCTF. A comparison between those platforms was conducted in [17], and analysing their results we chose to use the CTFd<sup>5</sup> framework. This platform allowed us to easily create, personalize, and manage the challenges associated with the tools to be tested, monitor the user progress and the status of the game, and to customise the interface.

Additionally, the success of the Jeopardy-style strategy relies on having clear and understandable challenges. Therefore, we defined a common structure to ensure a homogeneous and comprehensive description of each task the participants needed to address. Specifically, each challenge included the following 6 sections:

- (1) *Use Case*: the use case related to the challenge (among the ones introduced in Section 2.2);
- (2) *Tool*: the name of the tool to be used in the task, along with access details such as IP address and credentials;
- (3) *Description*: a brief description of the context to help the participant understand the relationship between the tool and the UC;
- (4) *Goal*: description of the actual task or issue to be addressed;
- (5) *Result*: a question that summarizes what the task is looking for;
- (6) *Answer Format*: specifications of the flag format to avoid misspelling errors (e.g., using capital letters, commas, dots, avoiding percentage symbols)

### 3.4 Evaluation Approach

To assess participants' engagement levels, evaluate the events, and enhance their experience for future events, two evaluation approaches were employed: one based on objective metrics and the other on learning schemes and quality measures. Both approaches were implemented during the second hackathon, whereas for the first event, our focus was solely on gathering objective metrics. This decision was made because, at that time, only a few tools were available, and their maturity level was not adequate for comprehensive evaluation. Recognizing these limitations, we did not include quality measurements in the evaluation of the first event.

<sup>5</sup><https://ctfd.io/>

**Objective metrics:** include information, statistics, and results that enable us to understand and evaluate the engagement level of participants and the status of the CTF competition. As discussed in Section 3.3, the management platform used allows us to directly extract and evaluate various statistics. These include the number of challenges attempted by each participant, their progression scores, challenges with the most attempts, overall comparison between attempts and solved challenges, and more. This information is processed post-event, providing insights into encountered difficulties and overall performance.

These metrics not only bridge the gap between participants' perceived learning experiences and their actual achievements but also enable organizers to identify objective weaknesses and errors that need to be addressed to improve user experience in future events.

**Quality measurements:** comprises information, feedback, and comments gathered during the event through a dedicated questionnaire. This survey is divided into three sections: an introductory section, an event evaluation section, and a tool evaluation section.

In the introductory section, general information about the attendees, such as their stakeholder group, role in the company/university, and previous participation in similar events, is collected. The second section consists of 8 questions: 6 are based on a 5-point Likert scale (ranging from 1 - Strongly Disagree to 5 - Strongly Agree), and 2 are open-ended questions. These questions evaluate the overall event, assess whether expectations were met, gather opinions about the CTF scenarios, and explore participants' interest in sharing the project goals beyond the event. Finally, the third section includes 8 questions structured according to the System Usability Scale (SUS) guidelines [22]. This section is designed specifically for tool evaluation, where each tool is rated on a 4-point scale from Unuseful (1) to Very Interesting (4). While the SUS guideline provides a 5-point Likert scale, we decided to omit one point to prevent a neutral position and encourage respondents to provide clear evaluations.

Although both evaluation approaches (objective metrics and quality measurements) are crucial for assessing the entire KINAITICS project, this work focuses exclusively on reporting and discussing objective metrics and the second section of quality measurements. Indeed, with this information, it will be possible to evaluate training impact and participant engagement across the hackathons. Yet, it is to be noted that, the evaluation of tools (third section of quality measurements) falls outside the scope of this research.

## 4 Experimental Framework

This section introduces the hackathon format, the gathered results, and the lessons learned. More specifically, Section 4.1 presents the hackathon agenda, while the results are analysed in Section 4.2. Finally, lessons learned are discussed in Section 4.3.

### 4.1 Hackathon Format

Both hackathons followed a similar agenda, composed of three parts: *Presentation*, *Hands-on*, and *Farewell*. In the first part, the organiser started with the presentation of the project in which the tools were developed, as well as the main goal of the hackathon event. Then,

the tool owners, called mentors, presented their tools in 10 minutes showing how they work through demos or live presentations.

In the second part of the event (*Hands-on*), the actual hackathon began. Indeed, participants started solving the challenges. Depending on the tool, the number of associated challenges and their difficulty levels varied, as shown in Table 1. In fact, in the first hackathon, there were 15 challenges to be addressed in two 1-hour sessions. In the second hackathon, 7 more tools were available, with 45 challenges. Therefore, three 1-hour sessions were organised.

**Table 1: Number of challenges per tool**

| #            | Tool           | # Challenges<br>First hackathon | # Challenges<br>Second hackathon |
|--------------|----------------|---------------------------------|----------------------------------|
| 1            | SeerBox        | 2                               | 2                                |
| 2            | NeuralSentinel | 6                               | 5                                |
| 3            | SED            | 6                               | 12                               |
| 4            | URAttack       | 1                               | 2                                |
| 5            | PenTestFox     | -                               | 4                                |
| 6            | MDPI           | -                               | 6                                |
| 7            | PhiShield      | -                               | 5                                |
| 8            | HDS            | -                               | 2                                |
| 9            | CyberShield    | -                               | 2                                |
| 10           | BOT            | -                               | 2                                |
| 11           | SHM            | -                               | 3                                |
| <b>total</b> |                | <b>15</b>                       | <b>45</b>                        |

Finally, in the third and final part of the hackathon (*Farewell*), the organiser concluded the session by announcing the winner and presenting the preliminary results of the event. It is to be noted that, an additional 10-minute session was introduced in the second event to gather feedback and results through a dedicated questionnaire. Indeed, during the first hackathon, we chose not to collect feedback due to the limited number of tools and their early development stage.

Both hackathons were organised as online events, the first hackathon took place on the 20th of July 2023 and lasted about 4 hours, while the second was on the 3rd of June 2024 with a duration of about 5 hours and 30 minutes.

### 4.2 Results

The results gathered from the hackathons are categorized into three main aspects: participant expertise, progress level (objective metrics), and insight and feedback (quality measurement).

**Participant expertise.** Hackathons were vibrant and dynamic events, marked by the enthusiastic participation of 11 competitors in the first event and 14 in the second. Furthermore, it should be noted that only one attendee participated in both events. This situation allowed us to gather objective feedback, avoiding bias and ensuring impartiality in the project results. The slight increase in the number of participants in the second hackathon was due

to enhanced efforts by the organizers, which included increased dissemination activities, promoting the event through social media and dedicated project-related forums, and offering small prizes for top scorers.

The participants came from various professional backgrounds within the cybersecurity field. They included researchers at different stages of their careers, such as assistants, post-doctoral researchers, and senior specialists, as shown in Figure 4a. This diverse group of experts worked collaboratively, leveraging their combined knowledge to tackle complex cybersecurity challenges.

In particular, Figure 4a shows that the majority of participants (6 out of 11) had a senior profile among cybersecurity specialists, security engineers, and post-doctoral researchers. On the other hand, the other 5 participants had junior profiles, with 4 being researchers. A different landscape is observed in Figure 4b for the second hackathon. This time, we had more junior profiles, 8 among researchers, master's, and PhD students, rather than senior profiles, e.g. 7. This mix of senior and junior talent in both events fostered a unique learning environment where experienced professionals and newcomers could collaborate and learn from each other.

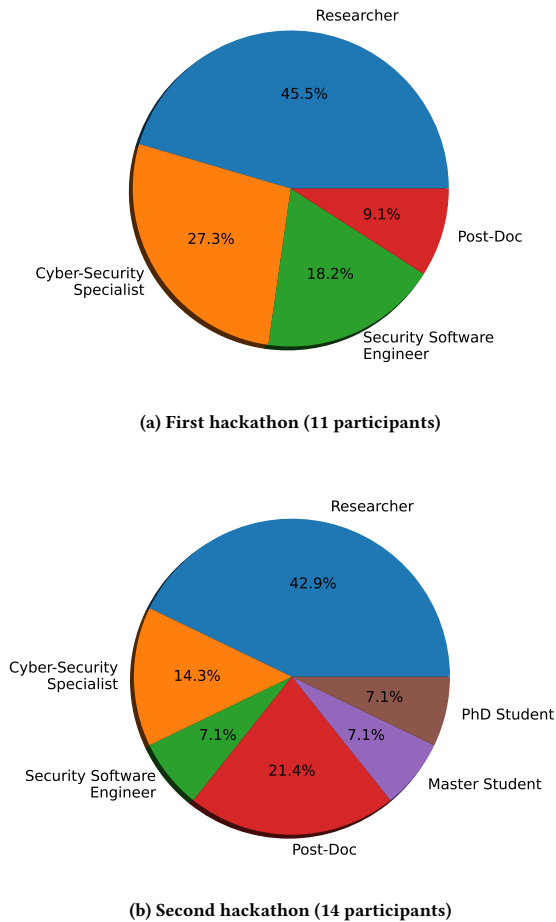


Figure 4: Expertise of the participants in both hackathons.

Figure 5 visually represents the distribution of nationalities among the participants, offering insights into the international makeup of the event. The first event saw participants from six European countries, with strong participation mainly from Italy and Spain. In contrast, the second event saw participation from just five European countries. The event lost participants from Austria and Portugal but saw an increase in participation from Belgium and Greece. Figure 5 illustrates how the hackathons brought together individuals from different parts of Europe, each contributing their unique perspectives and skills.

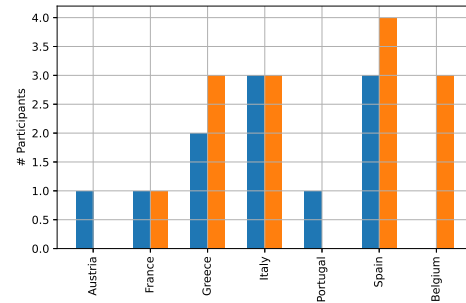


Figure 5: Nationality of the hackathon participants.

**Progress Level.** Table 2 compares various statistics and results of both hackathons. In particular, as previously mentioned, the second hackathon featured a significantly greater number of available tools, nearly triple the number compared to the first event. This increase/ratio was also reflected in the number of challenges, and consequently, in the maximum possible score. Furthermore, the table allows for an evaluation of the difficulty levels of both hackathons. Despite slight differences, the second event was more difficult than the first one, but overall, both were of medium difficulty. This consideration is supported by looking at the number of participants who reached at least half of the possible scores, which was  $\geq 57\%$  in both hackathons. This demonstrates that participants, although encountered some difficulty, were not completely stuck or frustrated in addressing the challenges. The same conclusion can be drawn from analysing the submission attempts. Yet, the attempts with correct answers were 61% and 55% in the first and second events, respectively, indicating that the challenges were neither too easy to be solved on the first attempt nor too hard, requiring too many attempts. Nevertheless, these results are outside from an ideal and aimed range of 65% and 75%. Finally, it should be noted that more than 59% of the challenges were successfully completed by at least half of the participants in both events. This demonstrates that participants, although encountering some difficulty, were not completely stuck or frustrated in addressing the challenges. Only a few challenges remained unsolved, suggesting a need to review their description and execution for future events.

**Insight and Feedback.** As discussed in Section 3.4, feedback and comments were gathered only in the second hackathon using a dedicated questionnaire. However, only 10 out of 15 hackathon participants completed the survey, and their answers are reported



**Table 2: Hackathon statistics and objective results.**

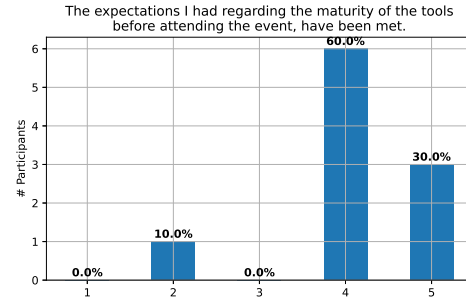
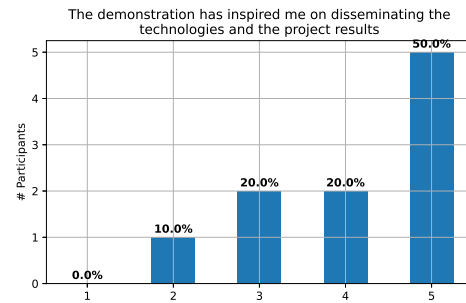
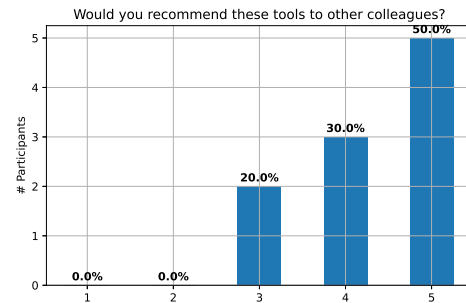
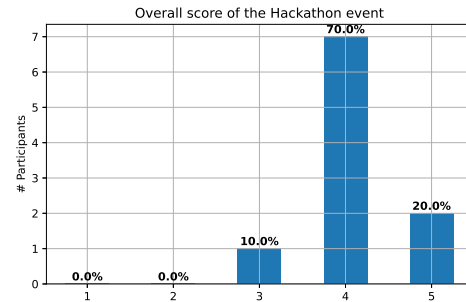
| Tool                                                 | First hackathon | Second hackathon |
|------------------------------------------------------|-----------------|------------------|
| Participants                                         | 11              | 14               |
| # Tools                                              | 4               | 11               |
| Challenges                                           | 17              | 45               |
| Maximum score available (max.score)                  | 850             | 2,525            |
| Top-score                                            | 650             | 1,625            |
| Participants with score $\geq$ max.score/2           | 63% (7)         | 57% (8)          |
| Submission % (Solved/Failure)                        | 61 / 59         | 55 / 45          |
| # solutions for the most solved challenge            | 7 (63%)         | 12 (86%)         |
| # challenges solved by at least half of participants | 10 (59%)        | 28 (62%)         |
| # challenges unsolved (no solution)                  | 2               | 3                |

in Figure 6. This figure reveals general satisfaction among the participants, their expectations, and what the project's current development. Specifically, 90% of the participants felt their expectations regarding the maturity of the tools were fulfilled (Figure 6a), while 70% expressed willingness to disseminate project results, with one participant dissenting, as shown in Figure 6b). Overall, the second hackathon was successful, as indicated by the overall score reported in Figure 6d.

Figure 7 reports comments from participants about their likes and dislikes regarding the event. Yet, these opinions confirm overall satisfaction and interest in such events, which provide a more enjoyable way to learn new cybersecurity deployments through hands-on experience. However, some users expressed concerns about the limited time available to address all challenges and test all tools. These aspects will be taken into account to improve their experience in future events.

### 4.3 Discussion and Lessons Learned

When reflecting on lessons learned and limitations encountered, it is important to differentiate between the first and second hackathon. In fact, following the first event, corrective actions were already implemented for the second. For instance, to mitigate connectivity issues with the landing hosts, we increased the number of machines and implemented monitoring to promptly detect issues, thereby avoiding balancing errors. This approach was crucial, as we did not encounter any disservice in the second event (as we had in the first). Additionally, to enhance user participation, organizers expanded event promotion across various communication channels and introduced rewards for the top scores. Although these efforts resulted in a slight increase in participant numbers, further improvements are necessary, such as selecting better event dates (avoiding summer periods) and expanding promotion in universities and cybersecurity forums.

**(a) Expectations before the event (1: Low - 5: High)****(b) The event has inspired you in disseminating the tools (1: Low - 5: High)****(c) Would you recommend the tool that was used? (1: Low - 5: High)****(d) Overall score of the event****Figure 6: Results gathered from the questionnaire shared after the second hackathon (1: Low - 5: High).**

|                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| What did you like/dislike about the demonstration?                                                                                                     |
| Creates a very clear view on how the tools work.                                                                                                       |
| The amount of time dedicated to PenTestFox was too little. We couldn't stabilize the shell to get the required info.                                   |
| All the tasks were really informative and interesting. I really liked the Healthcare use-cases.                                                        |
| Instructions and setup not entirely clear. Too many tools, each with different instructions/setup, majority of the hackathon was reading instructions. |
| I liked the way the different technologies were presented and the ease to ask the organizers when something was not working properly                   |
| hands on demonstration                                                                                                                                 |
| Some of the tools need a bit more of development                                                                                                       |
| I liked the way of teaching the different processes through games, making it more enjoyable and didactic.                                              |

**Figure 7: Feedback received by the participants.**

Another significant action was the introduction of a dedicated questionnaire to evaluate both tools and event organisation. This feedback highlighted that participants found the number of challenges and tools adequate, so we do not need to push these aspects. However, we need to better organise the sessions, to have a more clear training session. In this regard, our idea is to separate tools into groups. This would involve having separate presentations and hands-on sessions for each group during designated times, rather than overwhelming participants with all presentations at once. This approach aims to streamline participant actions and enhance their engagement levels.

## 5 Conclusions

This paper presents our training experiences and findings within the context of the KINAITICS H2020 project. It includes a detailed description of the infrastructure deployed in two hackathon events, decisions taken to enhance participant engagement, evaluates their impact on both events, and compares outcomes in terms of participants' expertise, progress, and feedback. Results show that following corrective measures, the second event saw a slight increase in participant numbers. Both events successfully balanced junior and senior cybersecurity researchers, creating an intricate learning environment where professionals at different stages of their careers could collaborate effectively. Engagement levels were also promising, with over half of the participants achieving at least half of the available points. Additionally, participants confirmed their interest and satisfaction with the dedicated questionnaire.

Despite the positive results obtained, a key lesson learned is that technical partners should dedicate more time to improving the maturity level of their tools and creating comprehensive user guides to support end users. Simultaneously, they need to review some challenges and try to reduce their complexity. At the project level, the coordinator keep working on trying to generate new ideas that foster cooperation and facilitate the sharing of knowledge and experiences among similar (or domain-related) projects. This approach aims to cultivate a more cohesive community, potentially attracting more hackathon participants and enhancing the impact of the outcomes achieved.

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